G Causal Mediation Analysis

In this section we include additional figures showing the average indirect effect (AIE) split up across tasks for GPT-J (Figure 18), as well as the AIE for other models we evaluated.

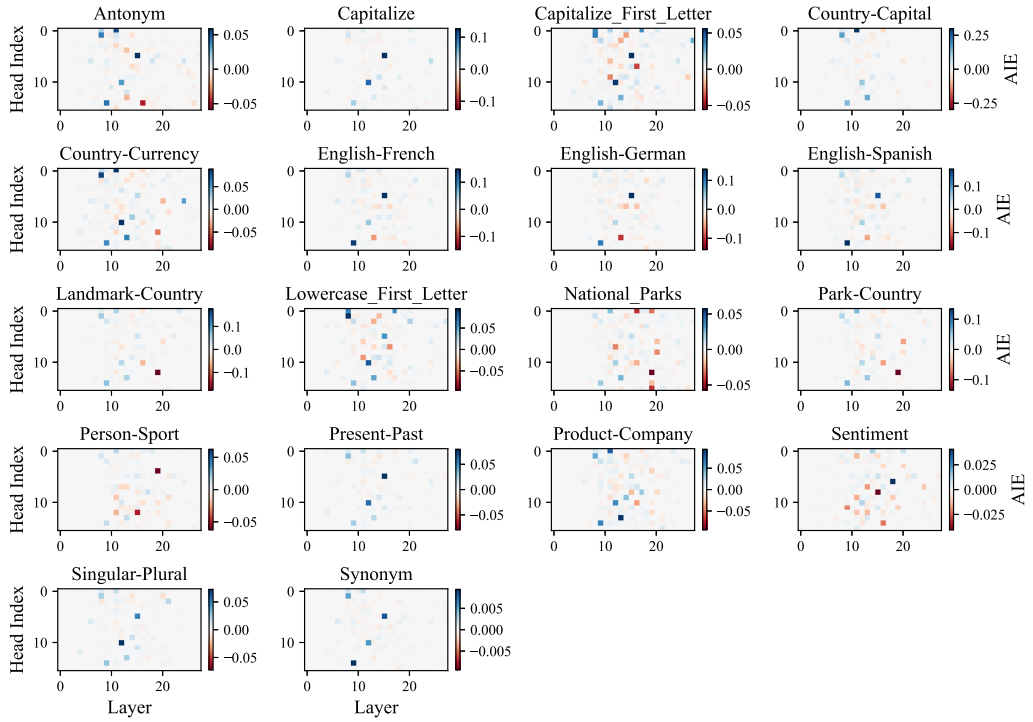

Figure 18: Average Indirect Effect (AIE) for attention heads in GPT-J, shown by task. The set of heads that have higher causal effect is fairly consistent across tasks.

The AIE for a particular model is computed across all abstractive tasks where the model can perform the task better than the majority label baseline given 10 ICL examples. We include heatmaps of the AIE for each attention head in Llama 2 (7B) (Figure 19), Llama 2 (13B) (Figure 20), and Llama 2 (70B) (Figure 22), as well as GPT-NeoX (Figure 21). For each model, the heads with the highest AIE are typically clustered in the middle layers of the network. In addition the maximum AIE across all heads tends to drop slightly as the size of the model increases, though the total number of heads also increases. In Llama 2 (7B) the max AIE is ≈ 0.047, while in Llama 2 (70B) the max AIE is ≈ 0.037.

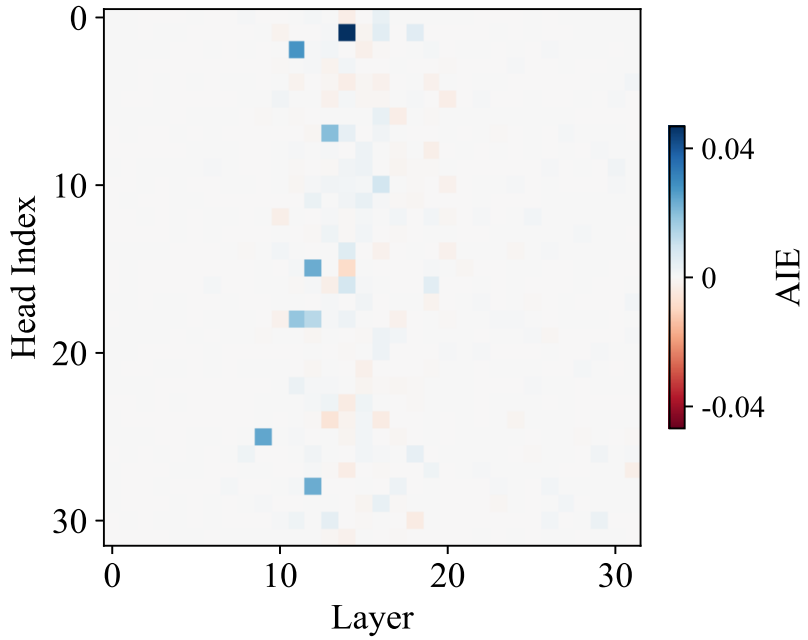

Figure 19: Average Indirect Effect (AIE) for each attention head in Llama 2 (7B) at the final token. This is computed across all abstractive tasks where the model can perform the task with 10 ICL examples. The heads with the highest AIE are mainly clustered in the middle layers of the network. Compared to GPT-J the AIE of the most influential heads is less (≈ 0.047 vs. ≈ 0.053 in GPT-J) but there are also more than double the number of attention heads

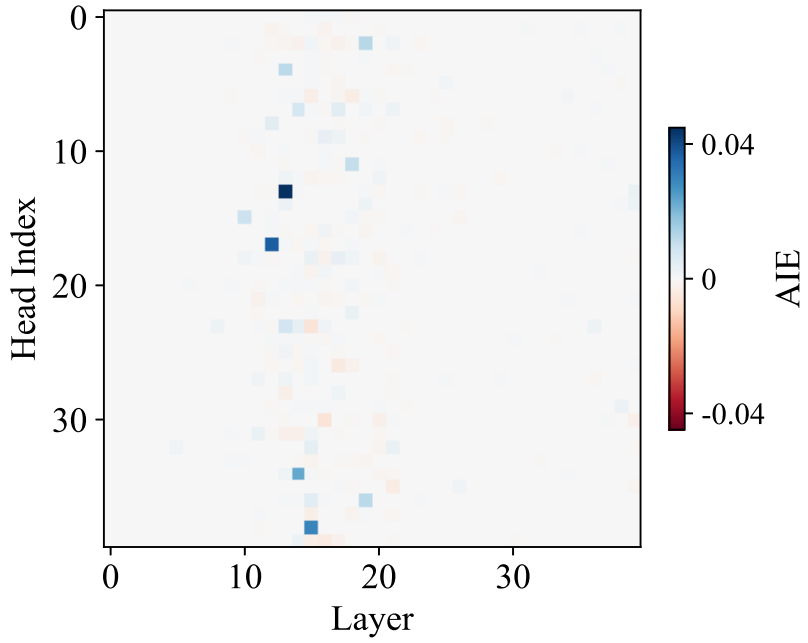

Figure 20: Average Indirect Effect (AIE) for each attention head in Llama 2 (13B) at the final token.

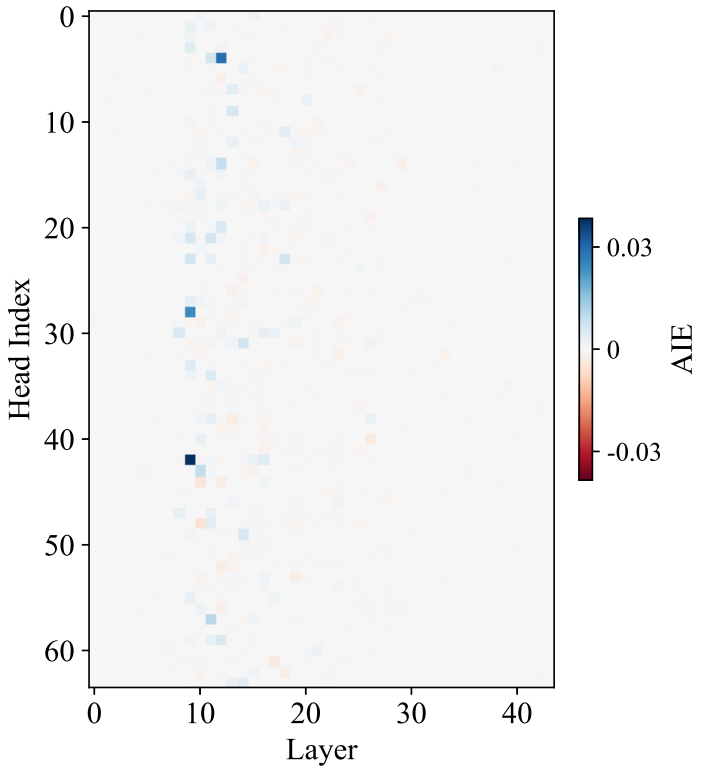

Figure 21: Average Indirect Effect (AIE) for each atttention head in GPT-NeoX at the final token. Interestingly, the heads with the highest AIE here are clustered in earlier middle layers (from layer 10-20), whereas in other models the heads are clustered more towards the middle of the network.

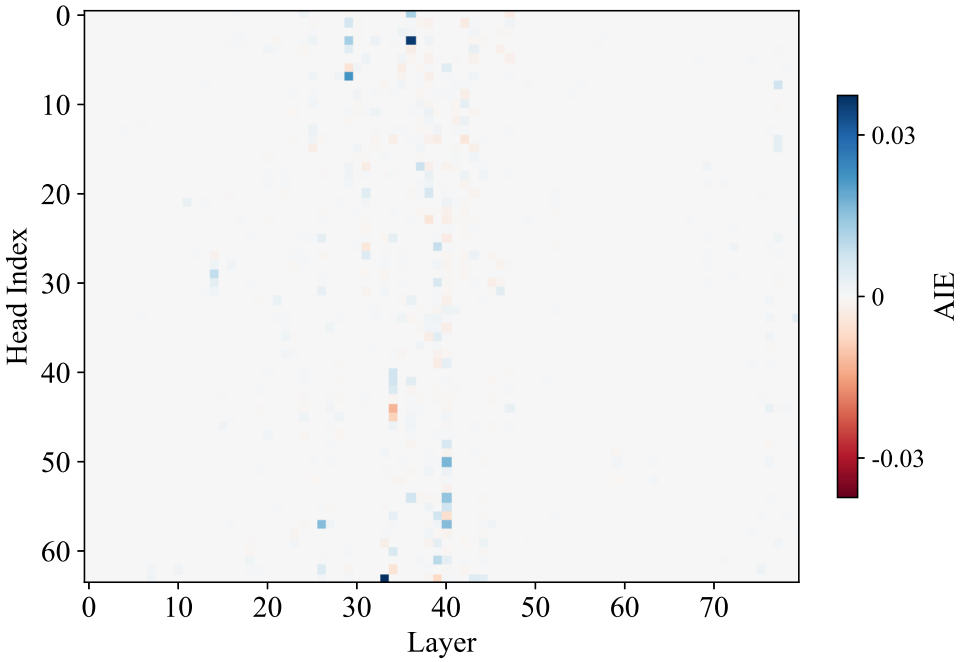

Figure 22: Average Indirect Effect (AIE) for each attention head in Llama 2 (70B) at the final token.